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أسئلة سنوات ميد لمادة الكيميا العامة ١

من بداية المادة إلى ما قبل درس

(موازنة المعادلات الكيميائية)

من ص ١ - ٧٤ من ملخص د.ريم

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the same value for their _____ quantum number.



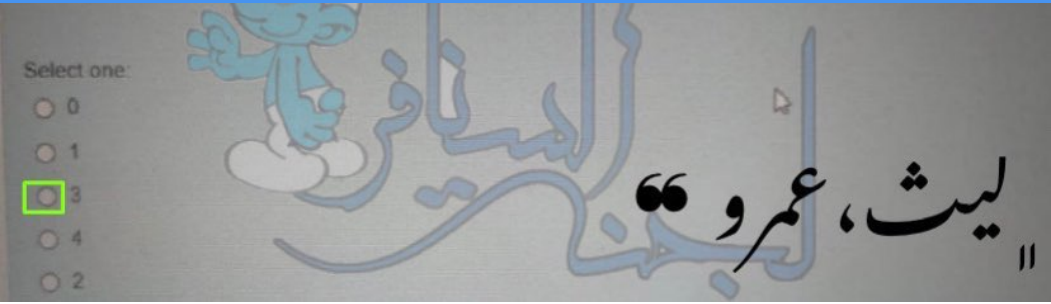
Select one:

- Secondary (Angular) (L)
- Spin (ms)
- Principal (n)
- Magnetic (mL)

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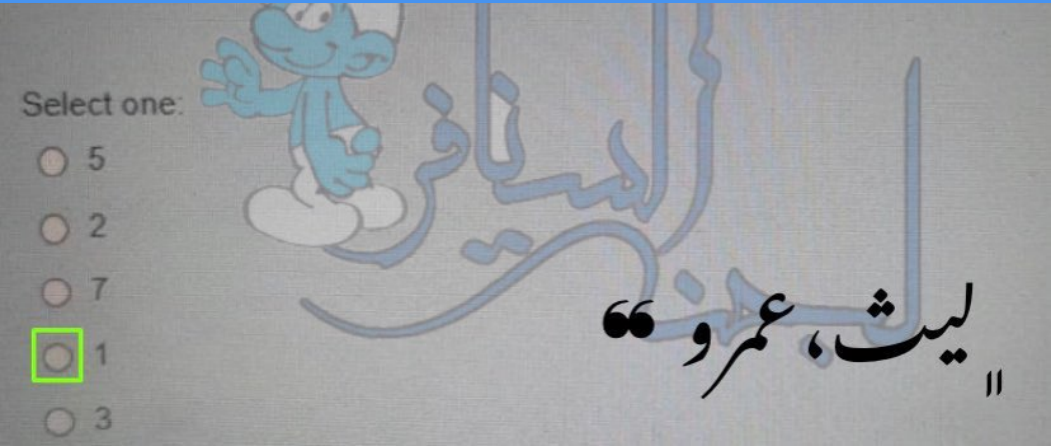
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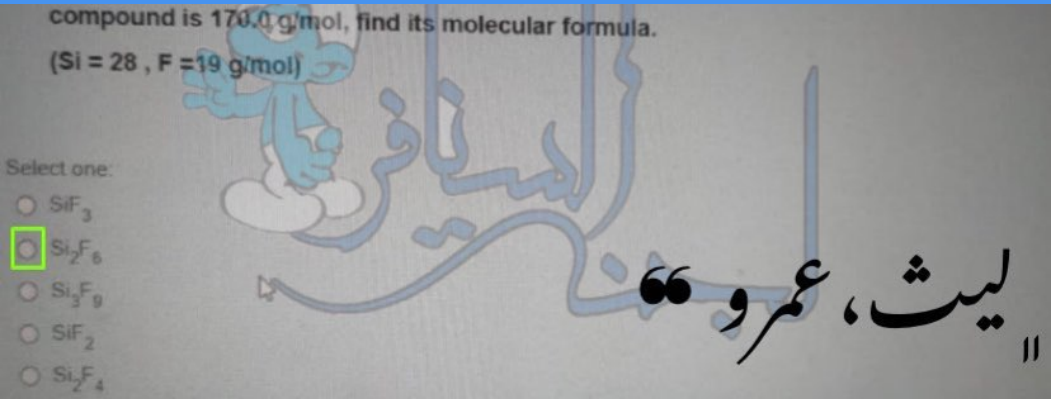
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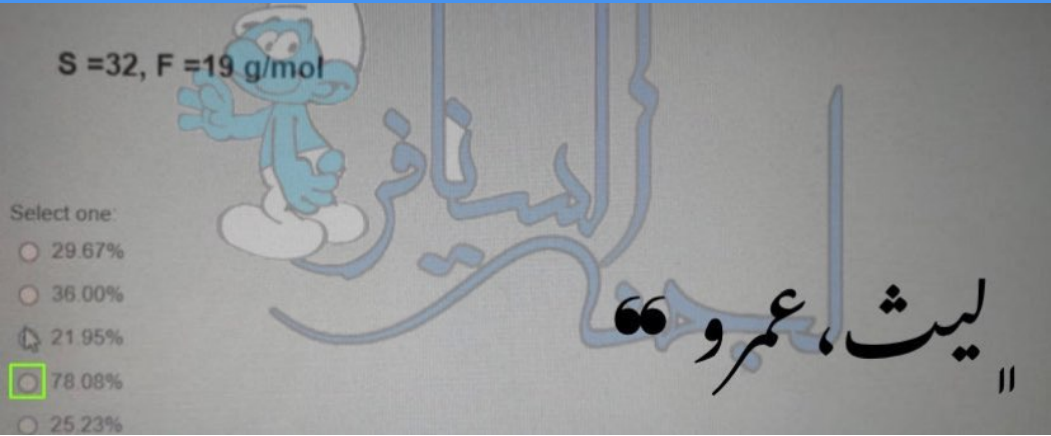
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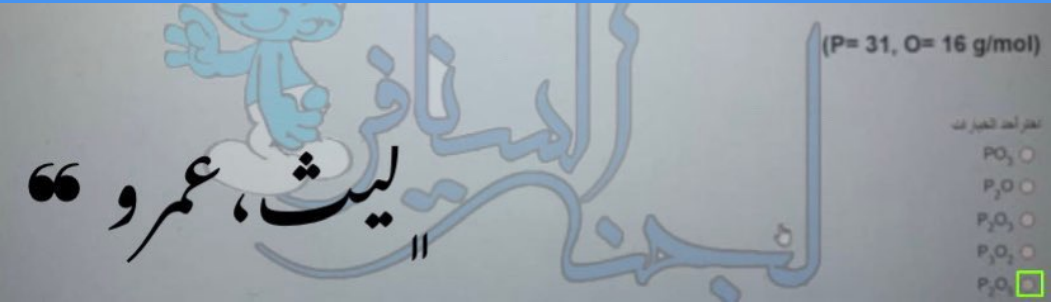
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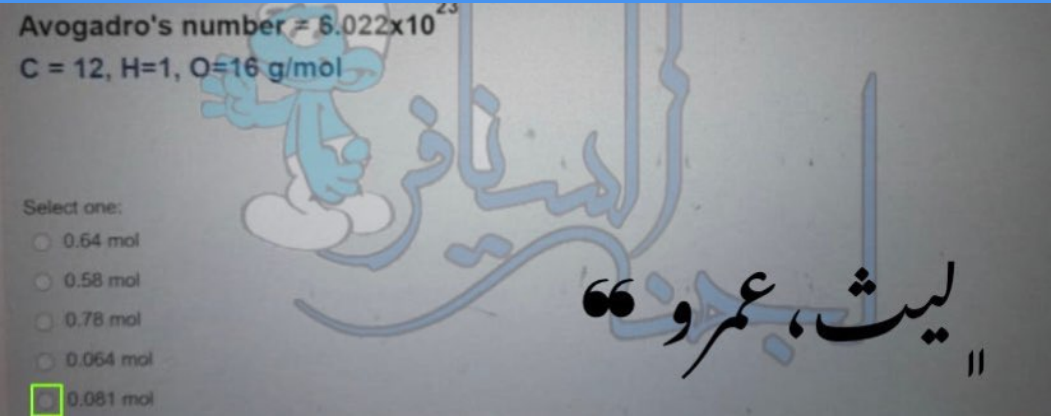
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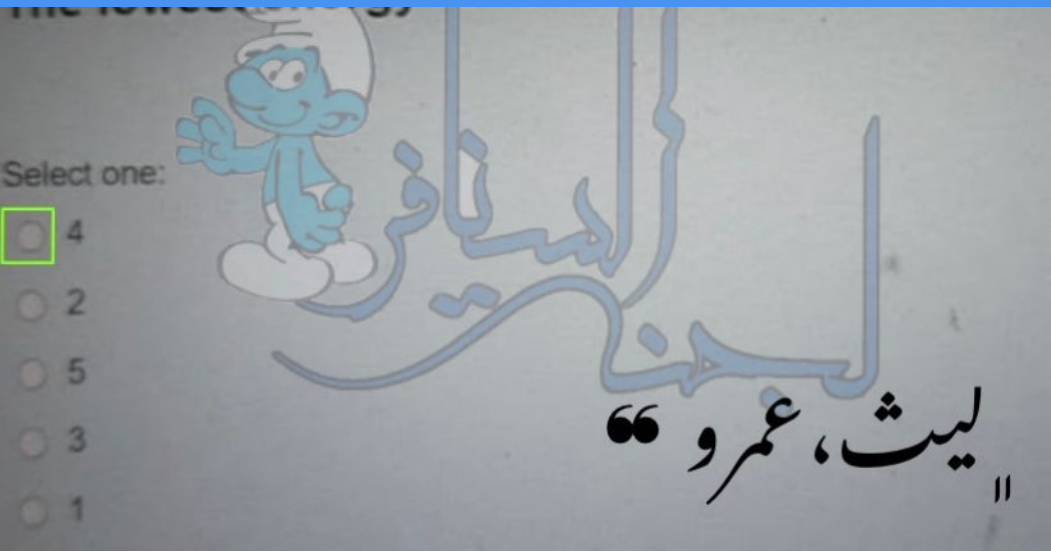
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(Si = 28 , F =19 g/mol)

Select one:

- Si_2F_6
- SiF_2
- SiF_3
- Si_2F_4
- Si_3F_9

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not an empirical formula?



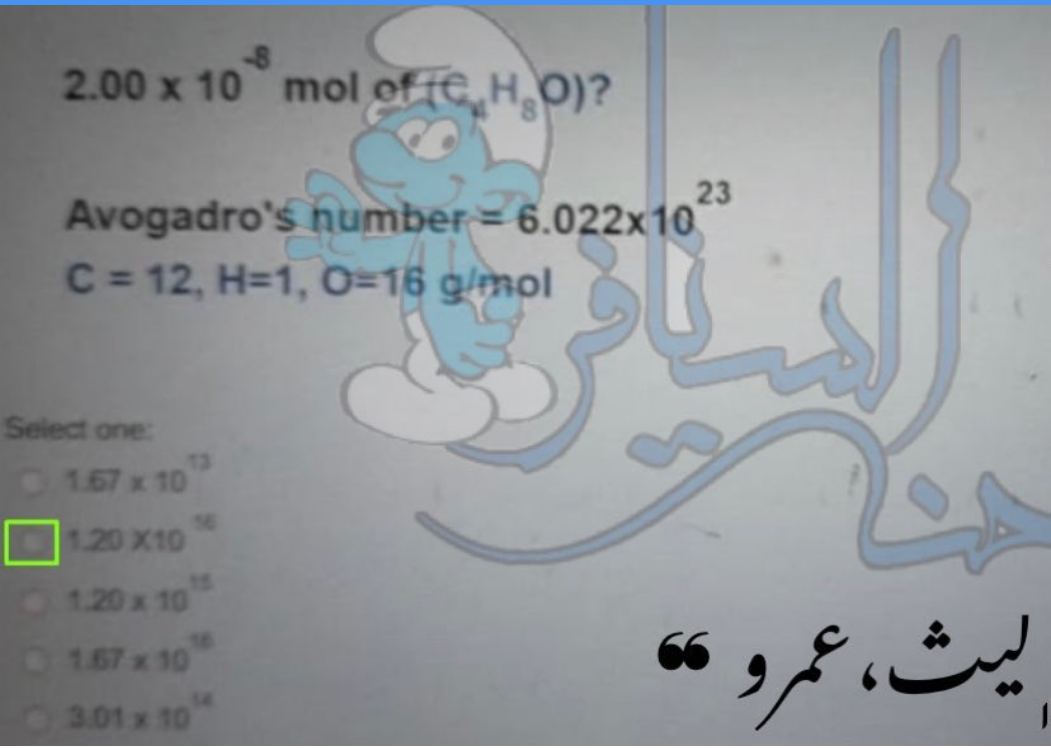
Select one:

- MoO_2Cl_2
- C_4H_8O
- $C_2N_2H_8$
- $BeCr_2O_7$
- Sb_2S_3

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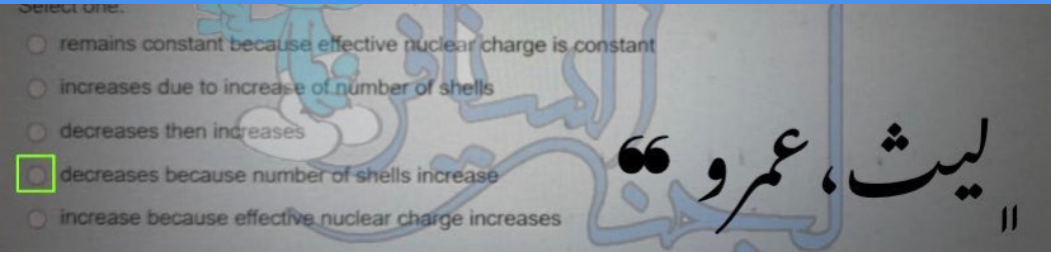
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(atomic number =14)

A. 2

B. 8

C. 7

D. 6

E. 4



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._____ (n) of the first p- orbital is

A. 1
B. 2
C. 3
D. 4
E. 0

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$S_4O_x = 224$ g/mol

Atomic weights: S = 32, 0 =16 g/mol

?What is the value x

A. 3

B. 5

C. 7

D. 4

E. 6

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ground state configuration for a

A. Transition element

B. Representative element

C. Halogen

D. Inner transition element

E. Nonmetal element

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an element that belongs to
,period 5

?group 2B in the periodic table

A. 47

B. 42

C. 41

D. 80

E. 48

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electron affinity (EA) for

$_{9}F$, $_{3}Li$, $_{10}Ne$

A. F

B. Li

C. Ne

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(Atomic weight of Ca = 40 g/mol)

A. 1.0 mole

B. 0.5 mole

C. 2.5 mole

D. 2.0 mole

E. 1.5 mole

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30) is a_____ element with unpaired electrons _____

A. paramagnetic , 3

B. paramagnetic , 2

C. paramagnetic, 6

D. diamagnetic , 2

E. diamagnetic , 0

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a nonmetal

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A) K (atomic no. = 19)

B) Mg (atomic no. = 12)

C) Li (atomic no. = 3)

D) Si (atomic no. = 14)

E) S (atomic no. = 16)

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of atoms) are found

in 7.5 mole of $K_2Cr_2O_7$

Avogadro's number = 6.022×10^{23}

[redacted]

4.698x10^25

A. 2.3×10^{25}

B. 7×10^{25}

C. [redacted] $\times 10^{25}$

D. 1.4×10^{26}

E. 3.6×10^{24}

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describes the energy of an orbital is:

A. Principle quantum number

B. Magnetic quantum number

C. Secondary (angular momentum) quantum number

D. Spin quantum number

E. None of the above

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A. increase, increase
B. increase, decrease
C. decrease, increase
D. decrease, decrease
E. is constant, decrease

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A. Principle quantum number
B. Magnetic quantum number
C. Secondary (angular momentum) quantum number
D. Spin quantum number
E. None of the above

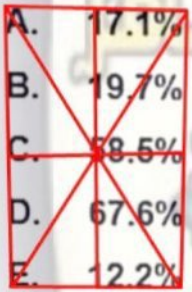
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of $Al(NO_3)_2$

(Atomic weights: Al= 27 , N=14, O =16 g/mol)



18.5%

الجواب 19.7

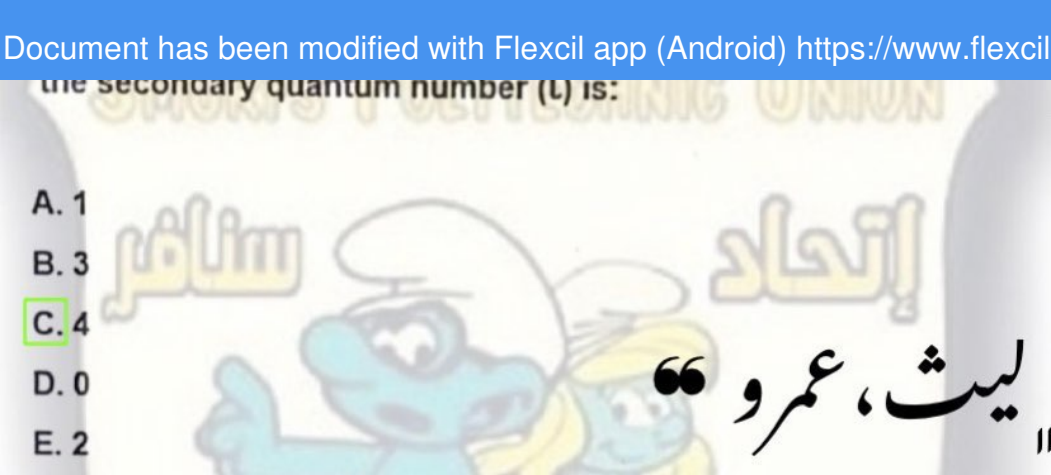
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the secondary quantum number (l) is:

A. 1
B. 3
C. 4
D. 0
E. 2



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atoms) are found

in 0.5 mole of $K_2S_4O_6$

Avogadro's number = 6.022×10^{23}

[redacted]

A. 2.3×10^{22}

B. 7×10^{25}

C. 5×10^{25}

D. 1.4×10^{26}

E. 3.6×10^{24}



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A. Ne

B. Ar

C. Li

D. Na



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quantum number set (n, l, m_l, m_s) equal to $(2,0,0, +\frac{1}{2})$

(Atomic no. of N = 7)

A. first electron

B. fourth electron

C. fifth electron

D. sixth electron

E. seventh electron

Third electron

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quantum number set (n, l, m_l, m_s) equal to (2,1,0, ½)

(Atomic no. of N = 7)

A. first electron

B. third electron

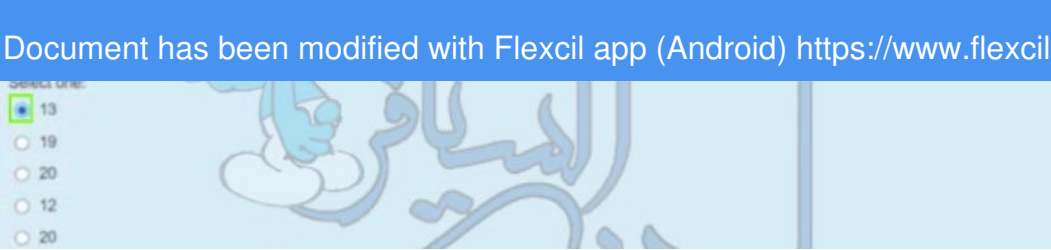
C. fifth electron

D. sixth electron

E. seventh electron

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compound that is contains

87.8 % of "C" and 12.2 % of "H"

(Atomic weights: C =12 g/mol, H= 1 g/mol)



A. C_2H_5

B. C_3H_8

C. C_2H_3

D. C_4H_{11}

E. C_3H_5

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metal

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A) Na (atomic no. = 11)
B) Si (atomic no. = 14)
C) B (atomic no. = 5)
D) Cl (atomic no. = 17)
E) Ar (atomic no. = 18)

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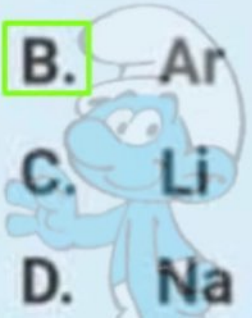
in period 3 ?

A. Ne

B. Ar

C. Li

D. Na



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is a_____ element

with _____ unpaired electrons

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A. paramagnetic , 3

B. paramagnetic , 5

C. paramagnetic , 6

D. diamagnetic , 2

E. diamagnetic , 0

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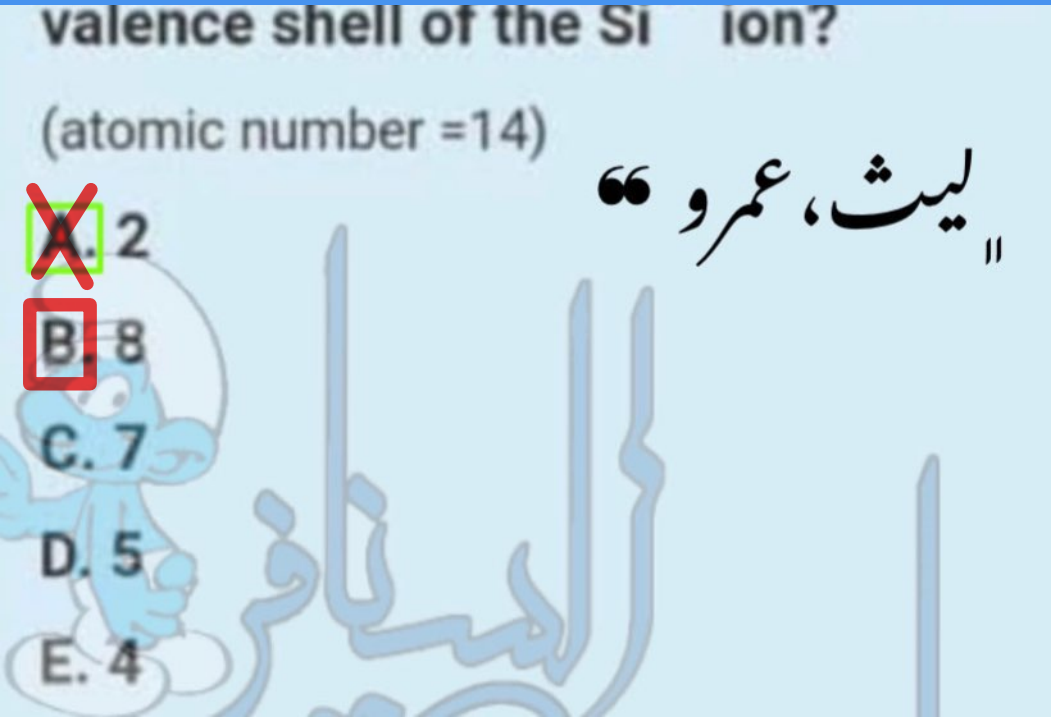
configuration for a

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A. Transition element
B. Alkali metal
C. Halogen
D. Inner transition element
E. Alkaline earth metal

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can be accommodated in a subshell for which $\ell = 4$ is:

s p d f g

0 1 2 3 4

Select one:

- 10
- 18
- 8
- 14
- 6



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configuration is _____.

Select one:

- $_{9}F^{-}$, $_{7}N^{2-}$
- $_{6}C^{2-}$, $_{8}O^{+}$
- $_{8}O^{2-}$, $_{11}Na^{+}$
- $_{13}Al^{2+}$, $_{12}Mg^{2+}$
- $_{10}Ne$, $_{9}F^{+}$



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number of an element that might have **seven** electrons in the **last *5d* subshell**?

Select one:

- 99
- 97
- 78
- 95
- 77



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configuration?

Select one:

- a. Se = [Ar]3d104s24p4
- b. Fe = [Ar]3d64s2
- c. H = 1s1
- d. O = [Ne]2s22p4



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$^{43}X^{2+}$ (Z = 21). The numbers of **neutrons** and **electrons** for this ion are, **respectively**:

Select one:

- 22 and 19
- 65 and 25
- 23 and 20
- 20 and 23
- 35 and 28



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(Atomic weight of Ca = 40 g/mol)

A. 1.0 mole

B. 0.5 mole

C. 2.5 mole

D. 2.0 mole

E. 1.5 mole



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of oxygen (O) in

of $Al(NO_3)_3$

(Atomic weights: Al =27 , N=14, O =16 g/mol)

A. 17.1%

B. 19.7%

C. 58.5%

D. 67.6%

E. 12.7 %

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atomic number for an element that belongs

to period 5, group 3B in the periodic table?

A. 39

B. 42

C. 41

D. 57

E. 48



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valence shell of the Si ion?

(atomic number =14)

A. 2

B. 8

C. 7

D. 5

E. 4



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Mn_2O_x = 174 g/mol

Atomic weights: Mn= 55, O = 16 g/mol

What is the value x?

A. 3

B. 5

C. 7

D. 4

E. 6



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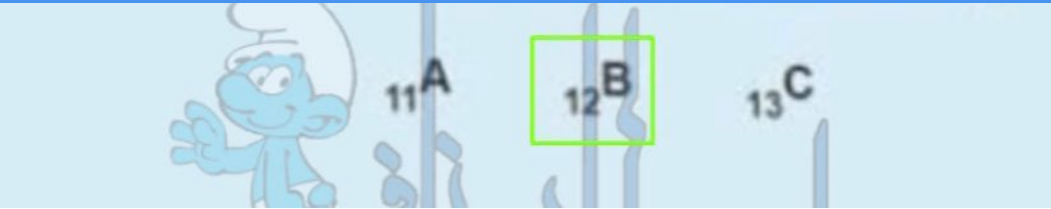
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has only one electron in its valence shell (outer shell) what is the atomic number of this element

Select one:

- 12
- 13
- 19
- 21
- 20



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electron affinity (EA) for

$_{9}F$, $_{3}Li$, $_{10}Ne$

A. F

B. Li

C. Ne

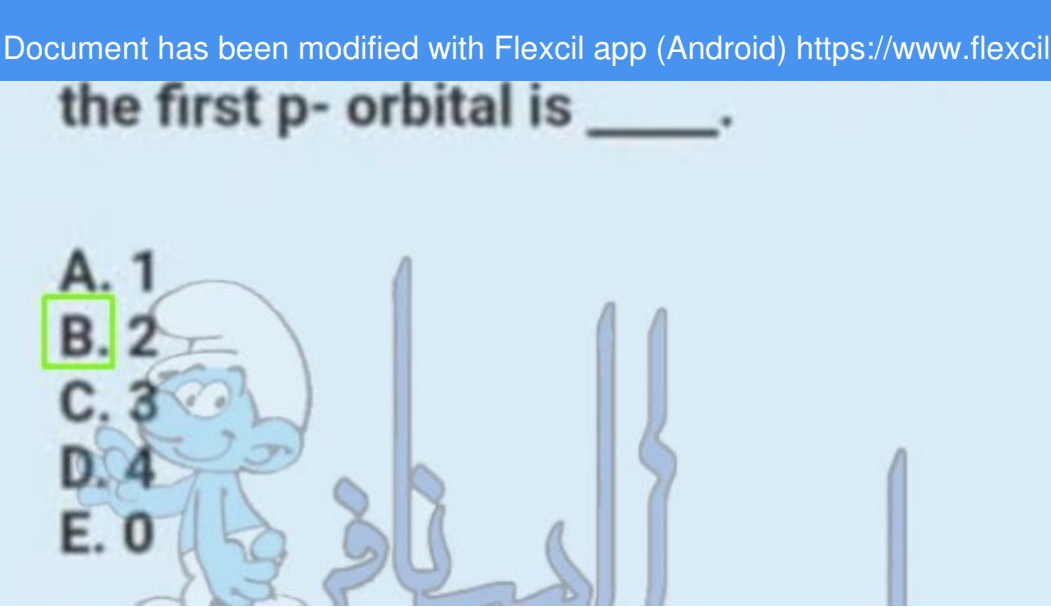


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the first p- orbital is _____.

A. 1
B. 2
C. 3
D. 4
E. 0



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Which of the following has the smallest radius?

- $^{+}Ca^{2}$
- ^{-}Br
- $^{-}S^{2}$
- Xe

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which has the following ground-state electron configuration?

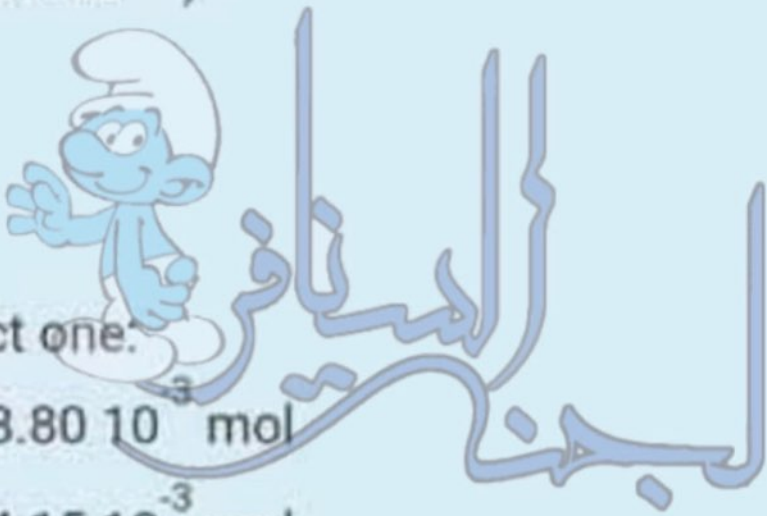
$[Ar]4s^2 3d^{10} 4p^5$

- Co
- Te
- Cl
- Ar
- None of them
- Br
- Se

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(Awt = 59 g.mol^{-1}) in a sample containing 5.0×10^{20} Co-atoms: (Avogadro's No. = 6.022×10^{23})



Select one:

- $3.80\ 10^{-3}$ mol
- $4.15\ 10^{-3}$ mol
- $6.30\ 10^{-4}$ mol
- $8.30\ 10^{-4}$ mol
- $4.15\ 10^{-4}$ mol

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N (Awt = 14 g.mol^{-1}) and 9.68% H (Awt = 1 g.mol^{-1}). What is the empirical formula for this compound is...

Select one:

- NH_2
- N_2H_2
- N_2H
- NH_3
- N_2H_3



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smallest diameter?

Select one:

- $_{31}Ga^{3+}$
- $_{16}S^{2-}$
- $_{13}Al^{3+}$ (Not Sure).
- $_{17}Cl^{-}$
- $_{19}K^{+}$

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A) K (atomic no. = 19)
B) Mg (atomic no. = 12)
C) Li (atomic no. = 3)
D) Cl (atomic no. = 17)
E) Ge (atomic no. = 32)

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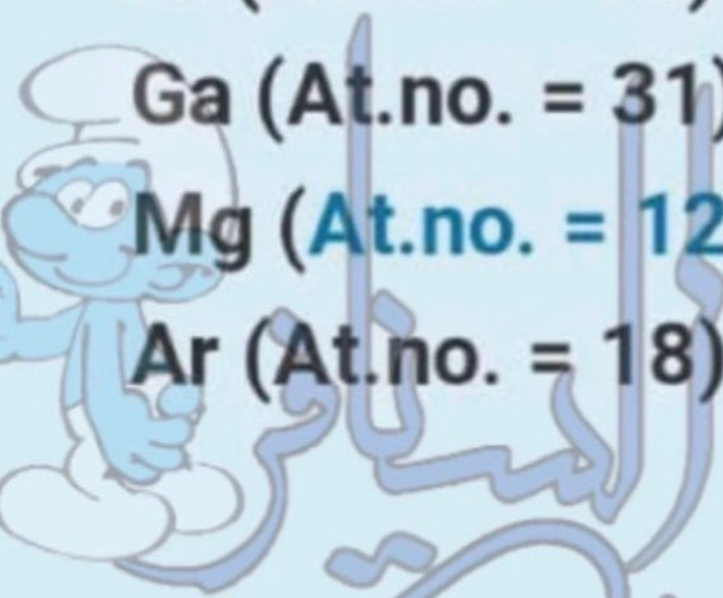
atoms according to decrease in their ionization energies

Al (At.no. = 13)

Ga (At.no. = 31)

Mg (At.no. = 12)

Ar (At.no. = 18)



Select one:

- Ar >Mg> Al > Ga
- Mg > Al> Ga > Ar
- Ga> Mg > Al > Ar
- [illegible] > Ga
- Al> Mg > Ar> Ga

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electronic configuration $[_{10}Ne]\ 3s^2\ 3p^6$ belongs to is:



Select one:

- IIB
- VIIIB
- VIIIA
- IIA
- VIA

"لیث، عمرو"

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D	C	A	D	C	B	B	C	A	E	D	C	E	D	A	E	C	D	C	A

1) The element $_{22}$**Ti** is a:
   a. diamagnetic representative element
   b. paramagnetic representative element.
   c. diamagnetic transition element.
   d. paramagnetic transition element.
   e. diamagnetic inner transition element.

2) Which of the following represents the **ground state** electronic configuration of chromium ($_{24}$**Cr**):
   Hint: Noble elements: $_{2}$**He**, $_{10}$**Ne**, $_{18}$**Ar**, $_{36}$**Kr**, $_{54}$**Xe**, $_{86}$**Rn**
   a. [Kr] $4s^2 3d^4$ b. [Ne] $3s^1 3d^5$ c. [Ar] $4s^1\ 3d^5$ d. [Kr] $5s^2 4d^4$ e. [Ar] $5s^2 4d^4$

3) One of the following sets of quantum numbers (**n, ℓ, $m_ℓ$**) describes the **last electron** in an **alkali metal** atom:
   a. (4,0,0) b. (3,1,0) c. (4,1,-1) d. (2,1,-1) e. (3,2,-1)

4) **$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$** is the ground state electronic configuration of an element that belongs to period _____, and group _____
   a. 4, 4A b. 4, 6A c. 4, 5A d. 4, 7A e. 4, 5B

5) One of the following is a ground state configuration of an element that belongs to the **actinide series**:
   a. [Xe] $6s^2\ 4f^{11}$ b. [Rn] $7s^2\ 5d^6$ c. [Rn] $7s^2\ 5f^7$ d. [Xe] $6s^2 5d^6$ e. [Kr] $6s^2 5f^6$

6) How many unpaired electrons does $_{27}Co^{2+}$ ion have?
   a. 2 b. 3 c. 5 d. 1 e. 0

7) An element ion having a **+4** charge has two electrons with n=1, eight electrons with n=2, and ten electrons with n=3. This element is:
   a. $_{25}$Mn b. $_{24}$Cr c. $_{26}$Fe d. $_{29}$Cu e. $_{23}$V

8) Which of the following statements about **oxygen ($_{8}$O)** and **fluorine ($_{9}$F)** is correct?
   a. Oxygen has smaller atomic size than flourine
   b. Oxygen has higher electron affinity than fluorine.
   c. Oxygen has lower first ionization energy than flourine
   d. Oxygen has more protons in its nucleus than flourine
   e .O^{2-} has smaller ionic size than F^-

9) How many orbitals are found in 4d subshell:
   a. 5 b. 4 c. 16 d. 7 e. 3

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d. Ar > Li > S > P > K e. K > P > S > Ar > Li

11) Arrange the above elements **(in question 10)** according to decrease in their **first ionization energy.**

a. Ar > Li > S > P > K b. K > Li > S > P > Ar c. Ar > K > P > S > Li
d. Ar > Li > P > S > K e. Ar > K > S > P > Li

12) What is the **molar mass** of barium perchlorate, $Ba(ClO_4)_2$?
(At.wts: Ba= 137.3, Cl = 35.5, O=16 g/mol)

a. 189.90 g/mol b. 239 g/mol c. 336.2 g/mol
d. 272.3 g/mol e. 304.3 g/mol

13) How many copper (Cu) atoms are found in a sample of copper weighing 6.93 g?
(At.wt of Cu = 63.5 g/mol) **(Avogadro's number = 6.022×10^{23})**

a. 9.17×10^{21} b. 0.917 c. 1.09×10^{22}
d. 0.0645 e. 6.56×10^{22}

14) A compound has a molecular formula of $C_{12}H_{24}O_{10}$. What is its empirical formula?

a. CHO b. C_2H_4O c. $C_4H_6O_2$ d. $C_6H_{12}O_5$ e. $C_2H_4O_3$

15) Calculate the **mass percent** of nitrogen **"N"** in 1.77 g $(NH_4)_2CO_3$.
(At.wts: N=14, C=12, O=16, H=1 g/mol).

a. 29.1 % b. 17.9 % c. 1.79 % d. 8.33 % e. 50.0 %

16) For a particular compound composed of C, H, and O. If 2.0 mol of it contains 1.2×10^{25} atoms of hydrogen, what is the **subscript** for the **hydrogen** atom in C_4H_7O ?

a. 5 b. 8 c. 6 d. 4 e. 10

17) Which of the following processes below represents the first ionization of an element "X"?

a. $X_{(g)} + e^- \rightarrow X^-_{(g)}$ b. $X_{(g)} \rightarrow X^{2+}_{(g)} + 2e^-$ c. $X_{(g)} \rightarrow X^+_{(g)} + e^-$
d. $X_{(g)} + 2e^- \rightarrow X^{2+}_{(g)}$ e. $X_{(g)} \rightarrow X^{2-}_{(g)} + 2e^-$

18) What decreases going from left to right (**within a period**) in the periodic table?

a. Electron affinity
b. Ionization energy
c. Number of shells
d. Atomic size
e. (a and b)

19) A compound that contains 72.5% of iron (Fe), and 27.5% of oxygen (O). Find the empirical formula of this compound. (At.wt Fe= 55.8, O =16 g/mol)

a. Fe_2O_5 b. Fe_2O c. Fe_3O_4 d. Fe_3O_2 e. Fe_2O_3

20) For a compound that has the empirical formula C_2H_3, and its molecular weight is equal to 129.6 g/mol. Find it's molecular formula.

a. $C_{10}H_{15}$ b. C_6H_9 c. C_4H_{12} d. $C_{14}H_{21}$ e. C_2H_3

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2) Which of the following is a ground state configuration of an element:

a. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}4p^6\ 5s^2\ 4d^7\ 5p^6$
b. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}4p^6\ 5s^24d^4$
c. $[Ar]\ 5s^2\ 4d\ ^{10}5p^3$
d. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^4$
e. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^4$

3) What is the atomic number for an alkali metal found in period 4?

a. 17 b. 20 c. 19 d. 12 e. 11

4) Which of the following sets of quantum number **(n, ℓ, $m_ℓ$)** describes an electron in the "d" subshell?

a. (4,3, -3) b. (3,0,1) c. (4,1,1) d. (2, 2, -1) e. (3,2,2)

5) How many unpaired electrons does Cr^{4+} have? (atomic number of Cr = 24)

a. 6 b. 4 c. 0 d. 2 e. 3

6) What happens to first ionization energy (**IE_1**) going from top to bottom down a group in the periodic table?

a. It decreases because the effective nuclear charge increase
b. It decreases because number of shells increase
c. It increases because number of shells increase
d. It decreases because number of shells decrease
e. It increases because effective nuclear charge decrease

7) Which of the following equations is **balanced?**

a. $10\ HSiCl_3 + 15\ H_2O \rightarrow H_{10}Si_{10}O_{15} + 30\ HCl$
b. $C_2H_4 + 2\ O_2 \rightarrow 2\ CO_2 + 2\ H_2O$
c. $FeBr_3 + 3\ H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 3\ HBr$
d. a and b
e. a, b and c

8) A compound of empirical formula (CH_5N). If its molar mass = 62 g/mol. What is its molecular formula?

a. $C_2H_{10}N_2$ B. $C_3H_{15}N_3$ c. CH_5N d. $C_4H_{20}N_4$ e. $C_5H_{25}N_5$

9) A compound of empirical formula ($C_4H_7O_2$). If mass percent of carbon "C" = 30.4 % and mass of hydrogen "H" = 8.9 %. What is the empirical formula of this compound?

a. C_2H_5O b. $C_2H_7O_3$ c. $C_3H_7O_4$ d. $C_3H_6O_2$ e. $C_4H_7O_2$

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a. 2900 b.17800 c. 4350 d.30000 e. none

IE2	1575
IE3	3220
IE4	?
IE5	16000
IE6	19800
IE7	22100

12) When 5.0 g of $KClO_3$ (122.5 g/mol) is heated, it decomposes to produce 1.5 g of O_2. What is the percentage yield of the reaction? $2KClO_3 \rightarrow 2KCl + 3O_2$

a. 76.6% b. 66.3% c. 86.7% d. 56.1% e. 81.6%

13) What is the mass percent of oxygen "O" in $Cr(NO_3)_3$?

a. 36.8% b. 54.5% c. 48.9% d. 41.9% e. 60.5%

14) How many water molecules would be found in 1.5 g of $MgSO_4.XH_2O$ (Molar mass of $MgSO_4.XH_2O$ = 246 g/mol)

a. 2.38×10^{22} b. 2.57×10^{22} c. 2.74×10^{22} d. 1.05×10^{22} e. 2.15×10^{22}

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Element	number
A	15
B	11
C	17
D	16
E	20
F	32
G	10

1) Which element is a halogen? C

2) Which has the highest IE_1 (of all elements)? G

3) Which has the lowest IE_1 of elements (A, B, C and D)? B

4) Which is has the highest metallic character of elements (A, B, C and D)? B

5) Which has highest electron affinity of elements (A, B, C and D)? C

6) Arrange atoms A, B, C, D, E, F according to decrease in their atomic size.

E > F > B > A > D > C

7) To which group and period does element "F" belong? Period 4 Group 4A

8) Which element has 3 unpaired electrons? A

9) Which element has a set of quantum number ($n=4$, $\ell=0$, $m_\ell=0$) for its last electron? E

10) Which is a diamagnetic element? E, G

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2) The number of H atoms in 0.05 mol of $NH_4H_2PO_4$ (molar mass 114 g/mol) is: (Avogadro's no. = 6.022×10^{23})
a) 2.17×10^{23} b) 2.53×10^{23} c) 2.89×10^{23} d) 1.45×10^{23} e) 1.81×10^{23}

3) An unknown element (X) has four energy levels, and is an alkali metal. What is the atomic number of (X)?
a. 35 b. 37 c. 38 d. 19 e. 20

4) How many electrons have quantum number $\ell=2$ in $_{28}Ni^{2+}$?
a. 5 b. 2 c. 8 d. 7 e. 6

5) Which of the following arrangements is correct concerning electron affinity?
a. $_{35}Br < _{17}Cl$ b. $_{9}F < _{3}Li$ c. $_{10}Ne > _{6}C$ d. $_{3}Li < _{11}Na$ e. none.

6) One of the following statements is correct:
a. Hund's rule states that maximum number of electrons occupied in an orbital equals two.
b. Atomic size increase as effective nuclear charge of atom increase.
c. In general, ionization energy decreases going from left to right in the periodic table.
d. Maximum number of orbitals in the fifth shell (n=5) equals 5.
e. The magnetic quantum number is related to the orientation of orbital in space.

7) Which set of three quantum numbers (n, l, m_l) corresponds to 4d orbital?
a) 4, 1, 1 b) 4, 3, 2 c) 4, 0, 0 d) 4, 2, 3 e) 4, 2, 0

8) Balance the following equation with the smallest whole number coefficients. What is the coefficient for $Al(NO_3)_3$ in the balanced equation?
$Al(NO_3)_3 + Na_2S \rightarrow Al_2S_3 + NaNO_3$
a. 2 b. 3 c. 1 d. 6 e.5

9) When 5.0 g of P_4 was allowed to react with 10.24 g of S_8, 11.3 g of P_4S_{10} was produced. Calculate percentage yield of this reaction. $4P_4 + 5S_8 \rightarrow 4P_4S_{10}$
(At. wts: P = 31, S = 32)
a. 79.6% b. 65.2% c. 55.6% d. 36.4% e. 96%

10) A 25.0 gram sample of a compound contains 6.64 g of potassium (K), 8.84 g of chromium (Cr) and 9.52 g of oxygen (O). Find the empirical formula of this compound. (At.wts: Cr = 52.0 , K=39.0 , O=16.0)
a. K_3CrO_3 b. K_2CrO_4 c. $K_2Cr_2O_7$ d. $KCrO_4$ e. $K_7Cr_2O_2$

11) Metal X combines with oxygen to form a compound with the formula X_2O_7. 0.0441 grams of oxygen (O) combines with 0.0432 grams of metal X. Calculate the atomic weight of X (in g/mol). (At. wt of O = 16.0)
a. 19.9 b. 4.48 c. 16.3 d. 54.9 e. 4.67

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D A A B D E C D C B A D E B E B E C C C E D D B A

1) Which of the following sets of quantum numbers is unacceptable for an electron in an $_{21}$**Sc** atom in its ground state?

a. n=1, ℓ= 0, m_ℓ= 0, m_s =1/2
b. n=2, ℓ= 0, m_ℓ= 0, m_s =1/2
c. n=3, ℓ= 1, m_ℓ= -1, m_s =1/2
d. n=3, ℓ= 2, m_ℓ= 0, m_s =1/2
e. n=3, ℓ= 0, m_ℓ= 0, m_s =1/2

2) Which of the following atomic numbers belongs to a diamagnetic element?

a. 12 b. 22 c. 20 d. 13 e. 14

3) Which of the following atoms has the smallest size?

a. $_{10}$Ne b. $_{19}$K c. $_{15}$P d. $_{18}$Ar e. $_{11}$Na

4) Aspartame, an artificial sweetener, has the molecular formula $C_{14}H_{18}N_2O_5$. What is the mass in grams of 1.16×10^{-2} moles of the compound? (Atomic weights: C = 12.0, H = 1.0, N = 14.0, O = 16.0 g/mol).

a. 2.6 b. 3.4 c. 0.3 d. 4.3 e. 5.0

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A	E	D&E	B	D	E	C	E	E	A	E	D	A	E	E	C	A	D	E	D

1) The element $_{38}Sr$ is a:
   a. diamagnetic representative element
   b. paramagnetic representative element.
   c. diamagnetic transition element.
   d. paramgnetic transition element.
   e. diamagnetic inner transition element.

2) Which of the following represents the **ground state** electronic configuration of chromium ($_{24}Cr$):
   Hint: Noble elements: $_{2}He$, $_{10}Ne$, $_{18}Ar$, $_{36}Kr$, $_{54}Xe$, $_{86}Rn$
   a. $[Ar]\ 4s^2 3d^4$ b. $[Ne]\ 4s^2 3d^4$ c. $[Kr]\ 4s^1\ 3d^5$ d. $[Kr]\ 5s^2 4d^4$ e. $[Ar]\ 4s^1 3d^5$

3) One of the following sets of quantum numbers (n, ℓ, m_ℓ) describes the **last electron** in a **noble** element:
   a. (4,2,0) b. (3,2,1) c. (4,0,0) d. (3,1,-1) e. (3,1,0)

4) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^4$ is the ground state electronic configuration of an element that belongs to period _____, and group _____
   a. 4, 4A b. 4, 6A c. 4, 5A d. 4, 7A e. 4, 5B

5) One of the following is a ground state configuration of an element that belongs to the **lanthanide series**:
   a. $[Xe]\ 6s^2\ 4d^7$ b. $[Rn]\ 7s^2\ 5d^6$ c. $[Rn]\ 7s^2\ 5f^7$ d. $[Xe]\ 6s^2 4f^8$ e. $[Kr]\ 5s^2 4f^8$

6) How many unpaired electrons does $_{30}Zn^{2+}$ ion have?
   a. 2 b. 3 c. 5 d. 1 e. 0

7) An element ion having a **+3** charge has two electrons with n=1, eight electrons with n=2, and thirteen electrons with n=3. This element is:
   a. $_{25}Mn$ b. $_{24}Cr$ c. $_{26}Fe$ d. $_{29}Cu$ e. $_{23}V$

8) Which of the following statements about **oxygen ($_8O$)** and **fluorine ($_9F$)** is correct?
   a. Oxygen has smaller atomic size than flourine
   b. Oxygen has higher electron affinity than fluorine.
   c. Oxygen has higher first ionization energy than flourine
   d. Oxygen has more protons in its nucleus than flourine
   e .O^{2-} has larger ionic size than F^-

9) How many orbitals are found in 5p subshell:
   a. 5 b. 4 c. 16 d. 7 e. 3

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a. K > P > S > Ar > Li b. K > Ar > S > P > Li c. Ar > K > P > S > Li
d. Ar > Li > S > P > K e. Li > P > S > Ar > K

11) Arrange the above elements **(in question 10)** according to decrease in their **first ionization energ**

a. Ar > K > S > P > K b. K > Li > S > P > Ar c. Ar > K > P > S > Li
d. Ar > Li > S > P > K e. Ar > Li > P > S > K

12) What is the **molar mass** of barium chlorate, $Ba(ClO_2)_2$?
(At.wts: Ba= 137.3, Cl = 35.5, O=16 g/mol)

a. 189.90 g/mol b. 239 g/mol c. 336.3 g/mol
d. 272.3 g/mol e. 304.3 g/mol

13) How many copper (Cu) atoms are found in a sample of copper weighing 0.97 g?
(At.wt of Cu = 63.5 g/mol) **(Avogadro's number = 6.022×10^{23})**

a. 9.17×10^{21} b. 0.917 c. 1.09×10^{22}
d. 0.109 e. 6.56×10^{22}

14) A compound has a molecular formula of $C_{12}H_{24}O_{18}$. What is its empirical formula?
a. CHO b. C_2H_4O c. $C_4H_8O_2$ d. $C_6H_{12}O_5$ e. $C_2H_4O_3$

15) Calculate the **mass percent** of oxygen **"O"** in 1.77 g $(NH_4)_2CO_3$.
(At.wts: N=14, C=12, O=16, H=1 g/mol).

a. 29.1 % b. 17.9 % c. 1.79 % d. 8.33 % e. 50.0 %

16) For a particular compound composed of C, H, and O. If 2.0 mol of it contains 7.22×10^{24} atoms of hydrogen, what is the **subscript** for the **hydrogen** atom in C_4H_7O ?
a. 5 b. 8 c. 6 d. 4 e. 10

17) Which of the following processes below represents the first electron affinity of an element "X"?
a. $X_{(g)} + e^- \rightarrow X^-_{(g)}$ b. $X_{(g)} \rightarrow X^{2+}_{(g)} + 2e^-$ c. $X_{(g)} \rightarrow X^+_{(g)} + e^-$
d. $X_{(g)} + 2e^- \rightarrow X^{2+}_{(g)}$ e. $X_{(g)} \rightarrow X^{2-}_{(g)} + 2e^-$

18) What decreases going from left to right (**within a period**) in the periodic table?
a. Electron affinity
b. Ionization energy
c. Number of shells
d. Metallic character
e. (a and b)

19) A compound that contains 70% of iron (Fe), and 30 % of oxygen (O). Find the empirical formula of this compound. (At.wt Fe= 55.8, O =16 g/mol)
a. Fe_2O_5 b. Fe_2O c. Fe_3O_4 d. Fe_3O_2 e. Fe_2O_3

20) For a compound that has the empirical formula C_2H_3, and its molecular weight is equal to 189 g/mol. Find it's molecular formula.
a. $C_{10}H_{15}$ b. C_6H_9 c. C_4H_{12} d. $C_{14}H_{21}$ e. C_2H_3

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